Distinct Values Subarrays II

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2428 | Sorting and Searching | https://cses.fi/problemset/task/2428

Given an array of n integers, your task is to calculate the number of subarrays that have at most k distinct values.

Input

The first input line has two integers n and k.

The next line has n integers $x_1,x_2,\ldots,x_n$: the contents of the array.

Output

Print one integer: the number of subarrays.

Constraints

- $1 \le k \le n \le 2 \cdot 10^5$
- $1 \le x_i \le 10^9$

Example

Input
5 2 1 2 3 1 1

Output
10